

Assignment - Mathematics - W/S - II

TOPIC: Polynomials

$$\begin{aligned} 1. a) & 5x^2 - 29x + 20 \\ &= 5x^2 - 25x - 4x + 20 \\ &= 5x(x-5) - 4(x-5) \\ &= (x-5)(5x-4) \text{ (Ans)} \end{aligned}$$

$\therefore$ Hence, the zeros are 5 and $\frac{4}{5}$.

Verification:

$$\left. \begin{aligned} \alpha + \beta &= 5 + \frac{4}{5} = \frac{25+4}{5} = \frac{29}{5} \\ \alpha\beta &= 5 \times \frac{4}{5} = 4 \end{aligned} \right\} \text{ (Ans)}$$

Now, $a=5$, $b=-29$ and $c=20$.

$$\left. \begin{aligned} \alpha + \beta &= \frac{-b}{a} = \frac{29}{5} \\ \alpha\beta &= \frac{c}{a} = \frac{20}{5} = 4 \end{aligned} \right\} \text{ (Ans)}$$

$\therefore$ Proved.

$$\begin{aligned} b) & x^2 - x - 72 \\ &= x^2 - 9x + 8x - 72 \\ &= x(x-9) + 8(x-9) \\ &= (x-9)(x+8) \text{ (Ans)} \end{aligned}$$

$\therefore$ Hence, the zeros are 9 and -8.

Verification:

$$\alpha + \beta = -8 + 9 = 1 \quad \text{ } \} \text{ (Ans)}$$

$$\alpha \beta = -8 \times 9 = -72$$

Now, $a = 1, b = -1$ and $c = -72$

$$\alpha + \beta = \frac{-b}{a} = 1$$

$$\alpha \beta = \frac{c}{a} = -72 \times 1 = -72 \quad \text{ } \} \text{ (Ans)}$$

$\therefore$ Proved.

c) $x^2 - 9$

$$\Rightarrow x^2 = 9$$

$$\Rightarrow x = \pm \sqrt{9}$$

$$\Rightarrow x = +3, -3$$

$\therefore$ Hence, the zeroes of the polynomial are 3 and -3.

Verification:

$$\alpha + \beta = 3 - 3 = 0 \quad \text{ } \} \text{ (Ans)}$$

$$\alpha \times \beta = 3 \times (-3) = -9$$

Now, $a = 1, b = 0, c = -9$

$$\alpha + \beta = \frac{-b}{a} = 0$$

$$\alpha \beta = \frac{c}{a} = -9$$

$$\text{ } \} \text{ (Ans)}$$

$\therefore$ Proved.

$$d) x^2 - 5x$$

$$\Rightarrow x(x-5) = 0$$

$\therefore$ Hence, the zeros are 0 and 5.

Verification:

$$\left. \begin{aligned} \alpha + \beta &= 0 + 5 = 5 \\ \alpha \times \beta &= 0 \end{aligned} \right\} \text{(ANS)} =$$

NOW, $a=1, b=-5, c=0$.

$$\left. \begin{aligned} \alpha + \beta &= \frac{-b}{a} = 5 \\ \alpha \beta &= \frac{c}{a} = 0 \end{aligned} \right\} \text{(ANS)} =$$

$$\therefore \text{Proved.}$$

$$e) x^2 - 2$$

$$\Rightarrow x^2 = 2$$

$$\Rightarrow x = \pm\sqrt{2}$$

$\therefore$ Hence, the zeros are $\sqrt{2}$ and $-\sqrt{2}$.

Verification:

$$\left. \begin{aligned} \alpha + \beta &= \sqrt{2} - \sqrt{2} = 0 \\ \alpha \beta &= \sqrt{2} \times (-\sqrt{2}) = -2 \end{aligned} \right\} \text{(ANS)} =$$

NOW, $a=1, b=0, c=-2$.

$$\left. \begin{aligned} \alpha + \beta &= \frac{-b}{a} = 0 \\ \alpha \beta &= \frac{c}{a} = -2 \end{aligned} \right\} \text{(ANS)} =$$

$\therefore$ Proved.

2: i) Given: zeros of $p(x) = \frac{3}{2}$ and $-\frac{1}{2}$.

Now,

$$\begin{aligned} p(x) &= K[x^2 - (\alpha + \beta)x + \alpha\beta] \\ &= K\left[x^2 - \left(\frac{3}{2} - \frac{1}{2}\right)x + \frac{3}{2} \times \left(-\frac{1}{2}\right)\right] \\ &= K\left[x^2 - x - \frac{3}{4}\right] \end{aligned}$$

Putting the value of constant (K) as 4.

$$\therefore p(x) = 4x^2 - 4x - 3 \quad (\underline{\text{Ans}})$$

ii) Given: zeros of $p(x) = \frac{1}{4}$ and $\frac{1}{2}$.

Now,

$$\begin{aligned} p(x) &= K[x^2 - (\alpha + \beta)x + \alpha\beta] \\ &= K\left[x^2 - \left(\frac{1}{4} + \frac{1}{2}\right)x + \left(\frac{1}{4} \times \frac{1}{2}\right)\right] \\ &= K\left[x^2 - \frac{3}{4}x + \frac{1}{8}\right] \end{aligned}$$

Putting the value of constant (K) as 8.

$$\therefore p(x) = 8x^2 - 6x + 1 \quad (\underline{\text{Ans}})$$

iii) Given: zeros of $p(x) = \frac{1}{a}$ and $\frac{1}{b}$.

Now,

$$\begin{aligned} p(x) &= K[x^2 - (\alpha + \beta)x + \alpha\beta] \\ &= K\left[x^2 - \left(\frac{1}{a} + \frac{1}{b}\right)x + \frac{1}{a} \times \frac{1}{b}\right] \\ &= K\left[x^2 - \frac{b+a}{ab}x + \frac{1}{ab}\right] \end{aligned}$$

Putting the value of constant (K) as ab .
 $\therefore p(x) = abx^2 - (b+a)x + 1$ (Ans).

iv) Given: zeros of $p(x) = -5$ and -3 .
Now,

$$\begin{aligned} p(x) &= K[x^2 - (\alpha + \beta)x + \alpha\beta] \\ &= K[x^2 - (-5 - 3)x + (-5) \times (-3)] \\ &= K[x^2 + 8x + 15] \end{aligned}$$

Putting the value of constant (K) as 1.
 $\therefore p(x) = x^2 + 8x + 15$ (Ans).

v) Given: zeros of $p(x) = 3$ and $-\frac{1}{5}$.
Now,

$$\begin{aligned} p(x) &= K[x^2 - (\alpha + \beta)x + \alpha\beta] \\ &= K[x^2 - (3 - \frac{1}{5})x + (3) \times (-\frac{1}{5})] \\ &= K[x^2 - \frac{14}{5}x - \frac{3}{5}] \end{aligned}$$

Putting the value of constant (K) as 5.
 $\therefore p(x) = 5x^2 - 14x - 3$ (Ans).

3. Given: $(x+1)$ is a factor of $p(x) = x^2 - 6x - 7$
Then, $p(-1) = (-1)^2 - 6(-1) - 7$
 $= 1 + 6 - 7 = 0$.

$\therefore$ Hence, -1 is a zero of $p(x)$ — (i)

$$\begin{aligned} \text{And } p(x) &= x^2 - 7x + x - 7 \\ &= x(x-7) + 1(x-7) \\ &= (x-7)(x+1) \end{aligned}$$

$\therefore$ Then, the other remaining zero of $p(x)$ is 7. — (ii)
(Ans).

$$\begin{aligned}
 4. \quad p(x) &= x^2 - 10x - 3000 \\
 &= x^2 - 60x + 50x - 3000 \\
 &= x(x - 60) + 50(x - 60) \\
 &= (x - 60)(x + 50) \text{ (Ans)}.
 \end{aligned}$$

$\therefore$ Hence, the zeros are 60 and -50.

5. Given: $p(x) = 2x^2 + 7ax - 15$ and one zero of $p(x) = 5$.

To find: Value of a .

$$\begin{aligned}
 \text{Now, } p(5) &= 2(5)^2 + 7a(5) - 15 = 0 \\
 &\Rightarrow 2 \times 25 + 35a - 15 = 0 \\
 &\Rightarrow 50 - 15 + 35a = 0 \\
 &\Rightarrow 35a = -35 \\
 &\Rightarrow a = -1 \text{ (Ans)}.
 \end{aligned}$$

$\therefore$ Value of a is -1.

$$\begin{array}{r}
 6. \quad x-2 \overline{) 2x^2 - 11x + 14} \quad [2x - 7 \\
 \underline{- 2x^2 + 4x} \\
 - 7x + 14 \\
 \underline{- (-7x + 14)} \\
 0
 \end{array}$$

$\therefore$ Yes, $(x - 2)$ is a zero of $p(x) = 2x^2 - 11x + 14$. (Ans).

7. Given: $(x-2)$ is a factor of $p(x) = x^2 - Kx + 10$
To find: value of K .

$$\text{Now, } p(2) = 2^2 - K(2) + 10 = 0$$

$$\Rightarrow 4 - 2K + 10 = 0$$

$$\Rightarrow 14 + 0 = 2K$$

$$\Rightarrow K = 7 \text{ (Ans.)}$$

$\therefore$ value of K is 7.

8. Given: (2) is a zero of $p(x) = 3x^2 - 17x + K$
To find: value of K

$$\text{Now, } p(2) = 3(2)^2 - 17(2) + K = 0$$

$$\Rightarrow 12 - 34 + K = 0$$

$$\Rightarrow K = 34 - 12$$

$$\Rightarrow K = 22 \text{ (Ans.)}$$

$\therefore$ value of K is 22.

9. To prove: $(x-2)$ is a factor of $4x^4 - 20x^3 + 23x^2 + 5x - 6$

Now, let, $q(x) \Rightarrow x - 2 = 0 \Rightarrow x = 2$, and

let $p(x)$ be $4x^4 - 20x^3 + 23x^2 + 5x - 6$

$$\text{Now, } p(2) \Rightarrow 4(2)^4 - 20(2)^3 + 23(2)^2 + 5(2) - 6 = 0$$

$$\Rightarrow 64 - 160 + 92 + 10 - 6 = 0$$

$$\Rightarrow -166 + 166 = 0$$

$$\Rightarrow 0 = 0$$

$\therefore$ Hence, proved (Ans)

that $(x-2)$ is a factor of $p(x)$.

10. i) The shapes of the poles shown are likely parabolic, representing quadratic polynomials. If the graph of a parabola opens downwards, the coefficient of x^2 (a) is negative, resulting in a maximum point at the vertex.

$$\begin{aligned} \text{ii) } p(x) &= 4\sqrt{3}x^2 + 5x - 2\sqrt{3} \\ &= 4\sqrt{3}x^2 + 8x - 3x - 2\sqrt{3} \\ &= 4\sqrt{3}x(x+2) - \sqrt{3} \end{aligned}$$

$$\begin{aligned} \text{ii) } p(x) &= 4\sqrt{3}x^2 + 5x - 2\sqrt{3} \\ &= 4\sqrt{3}x^2 + 8x - 3x - 2\sqrt{3} \\ &= 4x(\sqrt{3}x+2) - \sqrt{3}(\sqrt{3}x+2) \\ &= (\sqrt{3}x+2)(4x-\sqrt{3}) \text{ (Ans)} \end{aligned}$$

$\therefore$ Hence, the zeros of $p(x)$ are $-\frac{2}{\sqrt{3}}$ & $\frac{\sqrt{3}}{4}$.

iii) Given: zeros are $\frac{\sqrt{3}}{2}$ and -7 of $p(x)$.
Now,

$$\begin{aligned} p(x) &= K[x^2 - (\alpha + \beta)x + (\alpha\beta)] \\ &= K\left[x^2 - \left(\frac{\sqrt{3}}{2} - 7\right)x + \frac{\sqrt{3} \times (-7)}{2}\right] \\ &= K\left[x^2 - \frac{(\sqrt{3} - 14)x + (-7)\sqrt{3}}{2}\right] \end{aligned}$$

Putting the value of K as 2, we get,
 $p(x) = 2x^2 - (\sqrt{3} - 14)x - 7\sqrt{3} = 0$ (Ans).
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